The amount of change in frequency indicates the speed of the fluid. As heart pumps blood in the body, it can be used to study the characteristics of it, and hence to detect any cardiovascular ailment.

Ultrasound components and working

The main component of an ultrasound device is a transducer, which converts an electrical signal to a longitudinal acoustic wave that travels through the body of the patient. It also catches the echoes reflected back by various organs.

The transducer is used in echoscopic probes in a linear, sectorial or convex fashion to get different fields of view. For instance, an echo-scopic probe with linear array of transducers is used in situations where the window or area of access to the body is comparatively large.

The transducer is supplied with short-duration high-voltage pulses using a pulse generator. Compensating amplifiers are used to amplify the received weak echoes after reflection. Digital processors and control units are used to process various signals that are transmitted and various signals that are received. A display unit shows the received signal.

The transducer sends out acoustic pulses, and each pulse's transmitting time and receiving time is recorded by the transducer. As the received signals are the echoes that are reflected back from surfaces of the organs, or from different obstacles faced by the pulses, each echo varies in its intensity. This intensity of the echoes is displayed as a function of distance to reveal various shapes (organs).

As an acoustic pulse travels through different mediums it may get reflected, refracted, scattered, diffracted, absorbed or be interfered with. The reflection is necessary for ultrasonic imaging to work. As the pulse encounters different organs with different shapes and sizes, different densities, compressibility and different bulk modulus, these may weaken the echo, or may destroy the pulse completely.

The behaviour of pulses in various mediums has been studied, and it has been found that different targets reflect back different intensities of echoes. Based on this, different targets can be

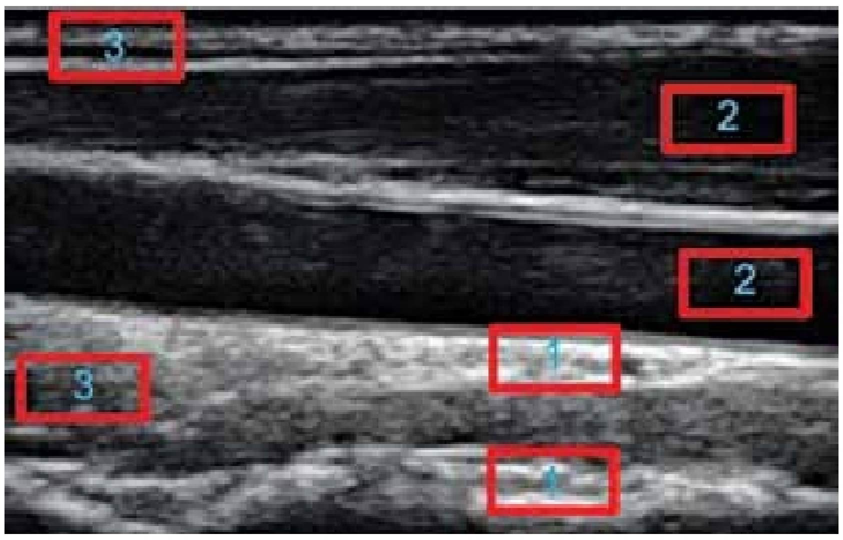

Fig. 1: Image showing different echo intensity areas

placed in three different categories:

1. Hyper-echoic
2. Hypo or anechoic
3. Iso-echoic

The hyper-echoic targets reflect most of the pulse energy that falls at their surface. This means they reflect back almost all of the pulses that collide with them. The brighter regions in a 2D ultrasonic image can be characterised by these targets. Typical examples of such targets are bones, air and capsules.

The hypo or anechoic targets do not reflect pulses at all. They absorb all the pulses that fall on their surface. The darkest regions in the display signify these targets. Some examples of these regions are body fluids such as blood. These areas cannot be seen through ultrasonic imaging.

The third category, iso-echoic, is of the targets that reflect back some amount of echoes. These neither fully absorb nor fully reflect back the incident pulses. These areas can be identified as normal intensity (not too dark, not too bright) on the display. Examples include body fat and muscles.

As air can also attenuate acoustic pulses, a gel is applied on the patient's body before ultrasonic imaging to minimise any trapped air between the probes and the body and maximise the output.

In Fig. 1, areas marked as 1 are hyper-echoic areas, those marked as 2 are hypo or anechoic areas, and areas marked as 3 are iso-echoic areas. Based on this information, we can presume that various targets on the display can be perceived through varying levels of intensities.

Although discrimination of an ultrasound equipment is very important, if two reflecting targets are very close to each other, the spatial resolution can play a very important role in recognising them as separate. In this regard, we can define two main types of spatial resolutions in ultrasound as (a) lateral resolution, and (b) axial resolution.

As mentioned earlier, the penetration of an acoustic beam depends on the frequency, density, compressibility and bulk modulus of the medium. At higher frequencies the diffraction and other losses decrease, but penetration also decreases. So, if density and compressibility of a medium is increased, it impedes the penetration of the wave. But if stiffness or bulk is increased, it facilitates better penetration of wave into that medium.

If two reflectors are parallel to each other and lie in direct scan line of ultrasonic wave, the minimum distance that can be recognised and differentiated is called the axial resolution or longitudinal resolution.

As attenuation,

$$\alpha(\text{Attenuation}) = 10\times\log_{10}\frac{\text{Echo intensity}}{\text{Propagated wave intensity}}$$

$$= 0.5(\text{dB cm}^{-1})\times 2\times d\times f$$

where d is the depth of the target that reflects back, f is the frequency (in MHz) and 0.5dBcm^{-1} is the assumed attenuation coefficient for travelling ultrasonic wave in soft tissue.

It can be seen that the attenuation is directly proportional to the depth of target reflector and frequency of the ultrasonic wave. This means that axial resolution is high when frequency of the acoustic wave is kept high and spatial pulse length (number of cycles in a pulse × λ (wavelength)) is kept short. The number of cycles depends on the damping nature of the transducer, which in turn depends on the geometry of it. So, in case of deep-target structures, low axial resolution is obtained.

When the target areas are perpendicular to the scanning ultrasonic wave, the distinguishing ability between two reflectors can be regarded as lateral resolution. This resolution